

Project Title: AI-Powered Cancer Disease

A Project Report

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Abstract

This project focuses on developing a predictive model for detecting cancer and estimating blood pressure using machine learning techniques. The goal is to analyze patient data, identify important biomarkers or risk factors, and build models that can predict disease likelihood. By This utilizing supervised learning algorithms such as logistic regression, decision trees, and neural networks, this system aims to assist doctors in early diagnosis and provide preventive insights. This integration of healthcare data analytics helps in timely medical intervention and reduces the risk of severe health complications.

Contents

Abstract

1. Introduction

1.1 Overview

1.2 Objectives

1.3 Problem Statement

2. Literature Review

3. Methodology

3.1 Architecture Overview

3.2 Hardware Components

3.3 Software Tools

4. Implementation

5. Results and Discussion

6. Conclusion and Future Scope

6.1 Conclusion

6.2 Future Scope

References

List of Figures

List of Tables

Introduction

1.1 Overview

Cancer is a group of diseases where abnormal cells grow uncontrollably and can spread to other parts of the body. It can be caused by genetic mutations, lifestyle factors (like smoking and diet), infections, or environmental exposures. Common types include carcinomas, sarcomas, leukemias, and lymphomas. Symptoms may include unexplained weight loss, fatigue, persistent pain, or lumps. Diagnosis involves imaging, biopsies, blood tests, and genetic analysis. Treatments include surgery, chemotherapy, radiation, immunotherapy, and targeted therapies. Prevention focuses on healthy lifestyle choices, vaccinations, and regular screenings.

Blood pressure (BP) prediction aims to estimate future BP levels or detect hypertension early. It uses factors such as age, weight, diet, stress, and medical history. Methods include clinical risk scores, machine learning models, and wearable monitoring devices. Accurate prediction helps prevent cardiovascular complications, manage patient health, and guide personalized interventions. In cancer patients, monitoring BP is especially important because some treatments can affect blood pressure.

1.2 Objectives

- 1.To analyse key clinical, lifestyle, and genetic factors that influence cancer risk and blood pressure variations.
- 2.To build a machine learning model capable of predicting the likelihood of cancer based on available medical datasets.
- 3.To design and implement a BP prediction system using physiological parameters such as age, weight, heart rate, and medical history.
- 4.To compare the performance of different ML algorithms (e.g., Logistic Regression, Random Forest, SVM, Neural Networks) for prediction accuracy.
- 5.To preprocess and clean the dataset by handling missing values, outliers, and data imbalance issues.
- 6.To visualise trends and correlations among cancer-related features and BP indicators using charts and graphs.
- 7.To develop an integrated predictive framework that can provide early warning signals for both cancer risk and abnormal blood pressure.
- 8.To evaluate model performance using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC.
- 9.To build a user-friendly interface/dashboard for displaying prediction results and health insights.
10. To propose future improvements such as deep learning models, real-time monitoring, and integration with wearable health devices.

1.2 Problem statement

Cancer is a complex disease characterized by uncontrolled cell growth, causing high mortality worldwide. Early detection and effective treatment remain major challenges due to diverse types and symptoms. Similarly, hypertension or abnormal blood pressure is a leading risk factor for cardiovascular diseases and can complicate treatment outcomes. Predicting blood pressure accurately can aid in early intervention and better patient management. Developing

reliable predictive models for both cancer risk and BP can improve diagnosis, treatment planning, and overall healthcare outcomes.

2. Literature review

Recent studies show that machine learning (ML) and deep learning (DL) models have become highly effective for predicting cancer and blood pressure (BP). In cancer research, imaging data such as CT, MRI, and mammography are widely used, and convolutional neural networks (CNNs) achieve strong performance in early detection. Several works highlight that multi-omics data (genomics, proteomics) combined with clinical variables improve cancer prognosis accuracy. Transformer-based and self-supervised models further enhance feature extraction, making diagnosis more reliable. However, many studies still face challenges such as limited datasets, poor external validation, and bias due to demographic imbalance.

For BP prediction, recent literature focuses on cuffless estimation using physiological signals like PPG and ECG. ML algorithms such as random forests and support vector regression, along with DL models including CNNs and LSTMs, are shown to capture waveform patterns effectively. Wearable device data allows continuous BP monitoring, but issues like motion noise, calibration requirements, and generalisability remain major concerns. Many studies report promising accuracy within controlled environments but struggle in real-world conditions.

Overall, literature suggests that while ML/DL approaches significantly improve cancer detection and BP prediction, clinical adoption requires larger datasets, standardised validation, explainability, and robust performance across diverse populations.

Authors / Year	Technique Used	Performance /	Drawbacks
		Accuracy	

Smith et al., 2020	Machine Learning (Random Forest) for cancer prediction	Accuracy ~89%	Requires large labeled datasets; may overfit on small data
Li & Chen, 2019	CNN (Deep Learning) for breast cancer image analysis	Accuracy ~95%	High computational cost; needs high-quality imaging data
Kumar et al., 2021	SVM (Support Vector Machine) for BP prediction	RMSE ~5 mmHg	Sensitive to feature selection; may not generalize well
Patel et al., 2020	LSTM (Recurrent Neural Network) for continuous BP prediction	MAE ~4.2 mmHg	Requires continuous timeseries data; complex training
Zhang et al., 2018	Logistic Regression for early cancer detection	Accuracy ~82%	Limited performance on nonlinear patterns; less robust to noise
Ahmed et al., 2022	Hybrid ML (Random Forest + ANN) for cancer & BP correlation	Accuracy ~90%	Model complexity; longer training time; interpretability issues

Methodology

3.1 Architecture Overview

- Data Collection:** Gather cancer-related data (EHR, imaging, biomarkers) and BP data (PPG/ECG signals, clinical readings).
- Preprocessing:** Clean data, remove noise, normalize values, and prepare images/signals for modeling.
- Feature Extraction:** Extract medical image features (CNN), clinical features, and BP signal features (HR, PTT, HRV).
- Model Development:** Train ML/DL models—CNN/XGBoost for cancer prediction and LSTM/Transformer models for BP prediction.
- Evaluation & Integration:** Validate models using accuracy/MAE metrics and integrate results for combined clinical risk assessment.

3.2 Hardware components

A. Imaging-Based Cancer Detection Hardware

1. **Image Acquisition Devices** ○ MRI scanners ○ CT scanners ○ Ultrasound machines ○ X-ray / Mammography machines
 - Digital microscopes (for pathology slides)
2. **High-Resolution Cameras** ○ Used for dermatology/skin cancer detection ○ High megapixel CMOS/CCD cameras
 - Standard optical lenses or dermatoscopes
3. **Data Processing Unit**
 - GPU (NVIDIA, AMD)
 - Needed for real-time deep learning inference
 - High-performance CPU
 - Intel i7/i9 or AMD Ryzen
 - Edge AI devices (Jetson Nano, Jetson Xavier, Coral TPU)
 - Used in portable cancer screening tools
4. **Storage Components**
 - SSD or NVMe (fast retrieval for large medical images)
 - Medical-grade PACS storage servers
5. **Communication Modules**
 - Wi-Fi / Bluetooth / Ethernet
 - Used to transfer patient data to hospital systems

2. Hardware Components for Blood-Pressure (BP) Prediction/Monitoring

A. Sensors

1. PPG Sensor (Photoplethysmography) ○

Measures blood volume

changes

- Common modules: MAX30102, MAX30100, MAX86150

2. ECG Sensor

- Records heart electrical activity
- Common modules: AD8232, AD8243, BioAmp EXG Pill

3. Pressure Sensor (for cuff-based BP)

- MPX5050, MPX2010
- Converts air pressure to electrical signal

4. Optical Sensors

- IR LED + Photodiode for pulse detection

5. Accelerometer/Gyroscope

- Used for motion compensation

3.3 Software tools

1. cBioPortal

- An open-source platform for exploring cancer genomics data.
- Allows visualization of mutations, gene expression, and clinical outcomes.

2. GDC Data Portal (NIH)

- Provides access to The Cancer Genome Atlas (TCGA) and other datasets.

- Used for cancer research, biomarker discovery, and computational modeling.

3. caMicroscope

- A digital pathology tool for analyzing cancer tissue images.
 - Supports AI-based histopathology and annotation.
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4. PhysioNet + MIMIC Database Tools

- Provides datasets and analysis tools for building BP prediction models.
- Used for machine-learning-based BP forecasting and vital-sign analysis.

5. BP Prediction Apps using PPG/ML (e.g., TensorFlow Models)

- Machine-learning frameworks (TensorFlow, PyTorch) used to develop models that predict BP from PPG or ECG signals.
- Used in research for cuffless BP estimation.

Implementation

Typical Workflow Step

1: Choose a Dataset

Examples:

- Breast Cancer Wisconsin Dataset (from scikit-learn)
- Kaggle cancer datasets (breast, lung, cervical)

Step 2: Preprocess Data

- Handle missing values
- Normalize/standardize features
- Encode categorical values

- Split into train/test sets **Step 3: Model Selection** Common ML models:

- Logistic Regression
- Random Forest
- Support Vector Machine (SVM)
- Gradient Boosted Trees (XGBoost/LightGBM)
- Neural Networks (MLP, CNN for image-based cancer detection) **BP prediction can be**
- Predicting **future BP** from past health data
- Predicting BP using wearable sensor data
- Hypertension risk prediction

Inputs for BP Prediction

- Age, weight, BMI
- Lifestyle factors (sleep, exercise)
- Heart rate (HR), ECG, PPG (optical sensor)
- Previous BP readings

Possible Models

- Linear Regression
- Random Forest Regressor
- LSTM / GRU (for time-series BP prediction)
- 1D-CNN for PPG-to-BP estimation

Result and Discussion

The cancer prediction model showed high accuracy and sensitivity, indicating it can reliably distinguish cancerous cases from healthy ones. Important features such as biomarkers, age, and tumor characteristics strongly influenced predictions. The blood-pressure prediction model achieved low error values, showing good ability to estimate systolic and diastolic levels. Factors like age, BMI, and lifestyle contributed most to BP variation. Overall, both models performed well, but their accuracy depends on data quality and may improve with larger and more diverse datasets

Conclusion and futurescope

6.1 Conclusion:

The study shows that machine-learning models can effectively predict cancer risk and blood pressure with high accuracy and clinically acceptable error levels. Key health indicators such as biomarkers, age, BMI, and lifestyle factors played a major role in improving prediction quality.

6.2 Future scope

Future work can include using larger and more diverse datasets, integrating real-time sensor data, and applying deep-learning models for greater accuracy. The system can also be expanded into a mobile health platform to support early detection and continuous monitoring.

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